

JMLR MLOSS reviewer comments and response status

kodama-cpp release candidate 0.1.0

Internal submission-readiness report

This report separates completed manuscript/repository revisions from evidence that cannot be represented as complete until the remaining benchmark matrix, tagged archive, and author declarations are available.

Comment	Response	Status
Legacy KNN comparison	A matched MetRef KNN benchmark compares the KNN-capable KODAMA 2.4.1 release with kodama-cpp single-core CPU, four-core CPU, and CUDA paths. It is not presented as the current CRAN baseline.	Completed on chiamaka
Current CRAN comparison	A separately labeled MetRef M=20/Tcycle=20 systems check compares KODAMA 3.3 CPU4 with kodama-cpp PLS-LDA CPU4. It reports unfavorable results unchanged and explicitly states classifier non-parity and the CUDA k limit.	Completed on chiamaka
Frozen public API	Version 0.1.0 macros, SemVer policy, CMake package version, wrapper versions, and a compile-link API snapshot test are present.	Completed locally
Version/tag/archive	Release checklist and clean-tag archive script prevent an uncommitted or dirty artifact from being cited; a DOI is optional.	External tag and deposit pending
License provenance	Pinned upstream snapshots, per-component licensing, SPDX tests, retained licenses, and the Metal Apache exception are documented.	Engineering audit complete; coauthor contribution confirmation remains
MLOSS page budget	The detailed manuscript is retained as a technical supplement; a separate official-style main paper targets four description pages plus references.	Completed; four description pages plus references verified
CUDA release build	Commit a32f718 passed all four configured license, numerical/core, public-API, and float32 smoke tests in the CUDA 13.0 build on chiamaka.	Completed on chiamaka
Continuous integration	GitHub Actions builds and tests the CPU core on Linux/macOS, native Metal on macOS, and the R/Python wrappers on Ubuntu. Separate coverage and API-documentation workflows passed for a32f718.	Completed at a32f718

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User documentation	A compact C++/R/Python walkthrough and generated Doxygen API are published through GitHub Pages.	Completed and URL verified
Active user community	The cover letter reports dated CRANlogs evidence for the predecessor R package and explicitly discloses that the new library has no public adoption metrics yet.	Independent kodama-cpp usage evidence pending
Python publication	The tested Python wrapper is included in the public core repository and exercised by CI.	Standalone public repository pending
Coverage depth	LLVM reports 63.58% line and 57.03% branch CPU coverage; hardware-specific tests cover Metal and manually validated CUDA paths.	Substantial state-machine, graph, PLS, and CUDA audit expansion pending
M/Tcycle rationale	Named MetRef/USPS sweeps report CV accuracy, ARI, active classes, runtime, and agreement-graph convergence; M=100 is justified by ensemble precision rather than external-label tuning.	Completed on CUDA
Final ablations	The release driver fixes M=100, Tcycle=100, landmarks, k, ncomp, splitting, graph correction, backend, wrappers, and external metrics before inspection.	Five-dataset CUDA panel complete; current-CRAN CPU4 check complete; exact current-CRAN CUDA match excluded because k=500 exceeds the native CUDA limit
Visualization claims	Classic and KODAMA UMAP/openTSNE use the same implementation/settings. The completed five-dataset panel retains positive and negative results, and the paper explicitly declines a universal improvement claim.	Completed for MetRef, PBMC3K PCA50, OptDigits, USPS, and Macosko2015 retina
Cover letter	A track-specific draft includes prior-publication disclosure, license, URL, version, software delta, and explicit author-only fields.	Draft complete; author fields pending
Release sustainability	README, changelog, contribution guide, API policy, prior-version delta, validation protocol, and release checklist are included.	Completed locally